

Cell Reports, Volume 17

Supplemental Information

Functional Mutations Form at CTCF-Cohesin

Binding Sites in Melanoma Due to Uneven

Nucleotide Excision Repair across the Motif

Rebecca C. Poulos, Julie A.I. Thoms, Yi Fang Guan, Ashwin Unnikrishnan, John E. Pimanda, and Jason W.H. Wong

Supplemental Data Items

Figure S1: CTCF motif, control motif and simulated mutations in skin cancers.

Figure S2: CTCF ChIP-seq and DNA-affinity experiments in COLO829.

Figure S3: Investigating enrichment for cancer-association in CTCF motif mutations in melanoma.

Table S1: CTCF and control motifs used in analyses.

Table S2: Accession numbers and cell-type information for RAD21 ChIP-seq datasets.

Table S3: COLO829 mutations in CTCF motifs identified from CTCF ChIP-seq in COLO829 cells.

Figure S1

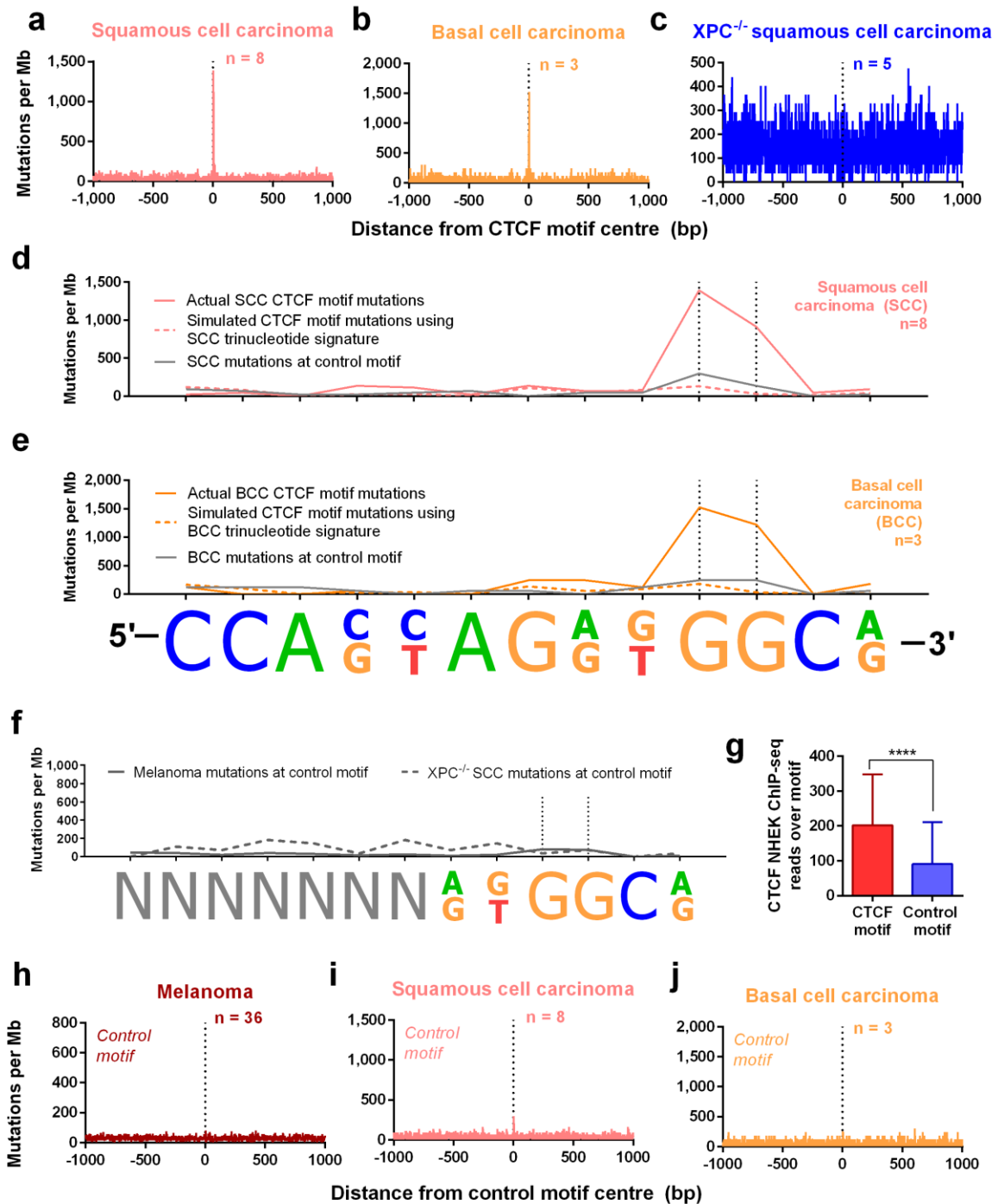


Figure S1: CTCF motif, control motif and simulated mutations in skin cancers. Related to Figure 1. Mutation density per sample for the region ± 1 kb of the CTCF motif centre in (a) skin squamous cell carcinoma (SCC; n=8), (b) skin basal cell carcinoma (BCC; n=3) and (c) nucleotide excision repair (NER)-deficient XPC^{-/-} SCCs (n=5). Mutation density per sample in (d) SCC and (e) BCC at the CTCF and control motifs, along with mutations simulated at CTCF motifs using the combined trinucleotide signature of SCC or BCC, respectively. (f) Mutation density per sample in melanoma and XPC^{-/-} SCC at control motifs. In the motif, 'N' denotes any nucleotide. (g) Number of reads from CTCF NHEK ChIP-seq across the strict 13 bp CTCF and control motifs (n=5,470), showing mean and standard deviation. Significance is by unpaired *t*-test, where **** *P* < 0.0001. Mutation density per sample for the region ± 1 kb of the control motif centre in (h) melanoma, (i) SCC and (j) BCC.

Figure S2

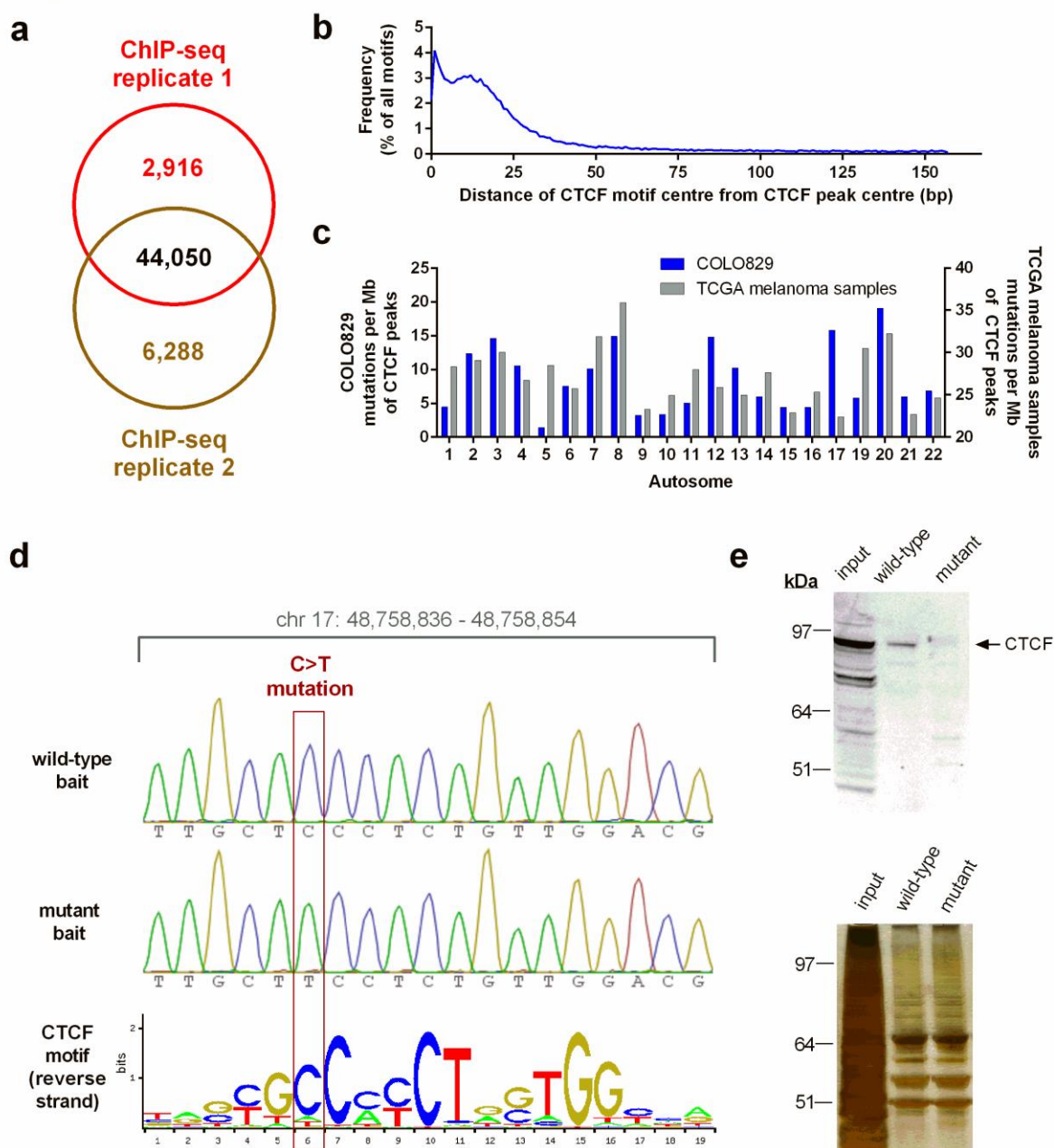


Figure S2: CTCF ChIP-seq and DNA-affinity experiments in COLO829. Related to Figure 3.

(a) Number of peaks identified in CTCF ChIP-seq replicates in COLO829. (b) Distance of the CTCF motif from the peak centre in COLO829 CTCF ChIP-seq data. (c) Distribution of CTCF mutations per Mb of CTCF peaks from COLO829 CTCF ChIP-seq in COLO829 and The Cancer Genome Atlas (TCGA) melanoma samples. (d) A portion of the Sanger sequencing traces from plasmids containing the cloned CTCF regions used to produce wild-type and mutant DNA baits, along with the reverse complement of the CTCF motif from the JASPAR database (Mathelier et al., 2015). (e) Western blot using anti-CTCF primary antibody (top) and silver stain loading control (bottom) from DNA affinity experiments. Blots show protein input (left lane), protein elute from wild-type DNA bait (middle lane) and protein elute from mutant DNA bait (right lane). Position of the CTCF protein band and protein molecular weight markers are indicated.

Figure S3

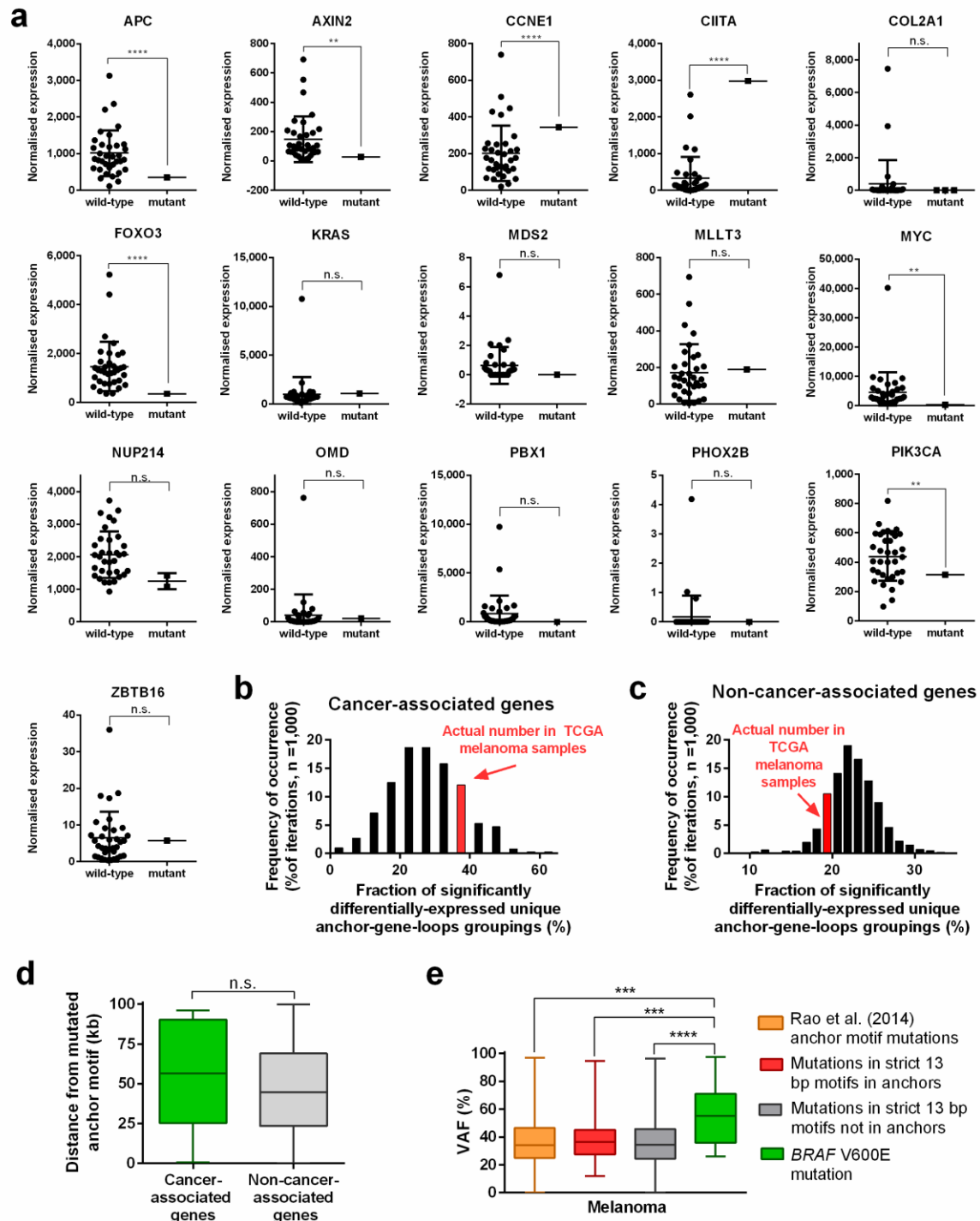


Figure S3: Investigating enrichment for cancer-association in CTCF motif mutations in melanoma. Related to Figure 4.

(a) Normalised gene expression (RNA-seq data) for melanoma samples, shown for all cancer-associated genes within loops with an anchor motif mutated in at least one melanoma sample. Adjusted P -values are given after multiple testing correction using the Bonferroni correction method. Error bars show mean and standard deviation, with individual data points indicating values for each sample. Histogram showing bootstrapping results ($n=1,000$ iterations) of the proportion of (b) cancer-associated genes (data in bins of 5% intervals) and (c) non-cancer-associated genes (data in bins of 1.25% intervals) with significantly different gene expression between wild-type and mutant samples for all unique anchor-gene-loop groupings when anchor motif mutations

were randomly assigned. The actual proportion of significant anchor-gene-loop groupings in The Cancer Genome Atlas (TCGA) melanomas is indicated. **(d)** Distance between the transcription start site of a given gene and an anchor motif, for all genes within 100 kb of an anchor motif mutated in melanoma, separated into all cancer-associated and non-cancer-associated genes. **(e)** Variant allele frequency (VAF) of all melanoma mutations within CTCF motifs in anchor regions, non-anchor regions and of *BRAF* V600E mutations, with only significant associations shown. Box-plots show median, quartiles, maximum and minimum. Significances are by unpaired *t*-test, or unpaired one-sample *t*-test where appropriate in **(a)**. **** $P < 0.0001$, *** $P < 0.001$, ** $P < 0.01$ and n.s. denotes not significant.

Table S1: CTCF and control motifs used in analyses. Related to Figure 1.

See Excel file.

Table S2: Accession numbers and cell-type information for RAD21 ChIP-seq datasets. Related to Figure 2.

ENCODE accession	Cell	Tissue	Karyotype	Mean peak size (bp)
ENCFF001VFE	GM12878	blood	normal	440
ENCFF001VRC	SK-N-SH	brain	cancer	1,045
ENCFF001VDS	A549	epithelium	cancer	311
ENCFF001VHB	H1-hESC	embryonic stem cell	normal	355
ENCFF001VLU	IMR-90	lung	normal	485
ENCFF001VJD	HeLa-S3	cervix	cancer	403
ENCFF001VOM	K562	blood	cancer	366
ENCFF001VKW	HepG2	liver	cancer	326

Table S3: COLO829 mutations in CTCF motifs identified from CTCF ChIP-seq in COLO829 cells. Related to Figure 3.

Position (hg19)	Mutation	Position in strict 13 bp motif	Position in 19 bp consensus motif	WGS reads		ChIP-seq reads		normVAF for mutant allele in CTCF ChIP-seq	Zygosity
				WT	MT	WT	MT		
chr20:49,435,093	G > A	-	2	17	11	53	46	57%	heterozygous
chr2:43,158,779	C > T	1	4	14	5	29	3	22%	heterozygous
chr3:196,336,325	C > A	2	5	8	15	138	2	1%	heterozygous
chr3:50,587,723	G > T	4	7	8	10	135	61	27%	heterozygous
chr8:43,265,125	G > A	4	7	17	16	18	0	0%	heterozygous
chr3:50,587,722	G > A	5	8	8	10	131	61	27%	heterozygous
chr2:233,304,367	C > T	10	13	8	5	190	8	6%	heterozygous
chr3:54,408,675	C > A	10	13	33	16	36	0	0%	heterozygous
chr7:150,685,763	C > T	10	13	13	7	149	37	32%	heterozygous
chr8:122,261,217	G > A	10	13	8	10	143	13	7%	heterozygous
chr17:10,231,077	G > A	10	13	11	25	60	80	37%	heterozygous
chr17:19,350,559	C > T	10	13	13	20	53	2	2%	heterozygous
chr17:19,350,558	C > T	11	14	15	20	53	2	3%	heterozygous
chr17:48,758,841	C > T	11	14	24	6	91	2	8%	heterozygous
chr19:43,389,232	G > A	-	18	10	23	53	128	51%	heterozygous
chr5:23,303,782	G > A	-	19	0	13	0	14	100%	homozygous

WGS = whole-genome sequencing; ChIP-seq = chromatin immunoprecipitation sequencing;

WT = wild-type; MT = mutant; normVAF = normalised variant allele frequency.

Supplemental Experimental Procedures

ChIP experiments in COLO829

ChIP-seq experiments were performed as previously described (Wilson et al., 2010). In brief, early passage cells were cross-linked in 1% formaldehyde and nuclei extracted and lysed. Chromatin was sonicated for 6 hours in a Diagenode (Wallonia, Belgium) Pico Bioruptor (30 seconds on/ 30 seconds off) and divided across two samples, incubated with polyclonal rabbit anti-CTCF antibody (Millipore 07-729) and rabbit immunoglobulin G (IgG; Sigma I5006) respectively. Immunoprecipitated DNA was purified and used for library construction, followed by sequencing using the Illumina HiSeq 2000 platform (BGI, Hong Kong). Two independent ChIP experiments using COLO829 cells grown to different passages and harvested independently were performed and sequenced separately. Replicate 1 was performed using ~20 million cells for each condition (CTCF and IgG) and replicate 2 was performed using ~40 million cells for each condition.

DNA-affinity experiments in COLO829

For DNA-affinity experiments, genomic DNA was isolated from COLO829 as previously described (Poulos et al., 2015) and a region (chr17:48,758,701-48,758,900) isolated for DNA baits using the following primers – forward: CTCCTTCACCGTTGTCTC; and reverse: CTGGAATGGAAGATGTGATG. The PCR product, representing both wild-type and mutant alleles, was cloned into the pGEM®-T Easy Vector System per manufacturer's instructions (Promega Corporation, WI, USA). Wild-type and mutant plasmid clones were selected and purified (**Figure S2d**). To produce wild-type and mutant DNA baits, the region was PCR amplified from wild-type and mutant plasmids respectively, using a 5' biotinylated forward primer so that each DNA bait had a biotin tag. PCR reactions were pooled to produce a sufficient quantity of DNA baits. The integrity of the PCR product was confirmed by Sanger sequencing.

1.5 µg of dsDNA baits were loaded with 300 µg of Streptavidin dynabeads (M-280; Thermo Fisher Scientific, NSW, Australia) and kept at room temperature for 3 hours. After incubation with a blocking buffer, DNA-conjugated beads were incubated with pre-cleared COLO829 cell lysate which had been extracted using the NE-PER™ Nuclear and Cytoplasmic Extraction Reagents per kit instructions (Thermo Fisher Scientific, NSW, Australia). Each 300 µg of DNA-conjugated bead condition was incubated with 0.2 mg of pre-cleared nuclear protein extract from COLO829, in the presence of 10 µM zinc from zinc acetate, overnight at 4°C. The bound proteins were boiled off the beads in 4x NuPAGE LD buffer (NP0007; Life Technologies, VIC, Australia) supplemented with 200 mM DTT at 95°C for 10 minutes. Equal volumes of elute from each DNA bait, and an aliquot of the protein input were loaded in separate lanes and separated on a 4-12% Bis-Tris gel (Life Technologies, VIC, Australia), and then transferred onto a nitrocellulose membrane. Western blotting was performed using 5% skim milk as a blocking reagent, polyclonal rabbit anti-CTCF antibody (Millipore 07-729) as the primary antibody and polyclonal goat anti-rabbit IgG (Dako P0448) as the secondary antibody. A silver stain was performed on the gel following transfer to the nitrocellulose membrane to serve as a loading control (**Figure S2e**). The silver stain was performed using reagents from Pierce Silver Stain for Mass Spectrometry per kit instructions (Thermo Fisher Scientific, NSW, Australia).

Variant allele frequency and copy number variation analysis

Variant allele frequencies (VAFs) in the melanoma samples were calculated by determining base calls at specific sites using SAMtools (Li et al., 2009) mpileup on the BAM files obtained from CGHub (Wilks et al., 2014). Reads were disregarded if they recorded a base call which was neither the wild-type nor mutant nucleotide (per the mutation calls made by Strelka (Saunders et al., 2012)). The VAF was calculated as the proportion of mutant base calls, over the total number of mutant and wild-type base calls, from all reads covering a given nucleotide.

Copy number segment information data for melanoma sample TCGA-D9-A148 was obtained from the Genomic Data Commons (GDC).

ChIP-sequencing data analysis

The NHEK CTCF ChIP-seq peaks used for analyses in this study were those from the processed 'narrow peak' BED files obtained directly from ENCODE. Peaks from the two replicates were merged to provide a complete list. To ensure that all regions used were of the same size, for all regions greater than 150 bp after merging, the centre 150 bp was used. To determine the CTCF read count from ChIP-seq across a given region, the

corresponding BAM files were used. BEDtools (Quinlan and Hall, 2010) was used to convert the BAM to BED files and extend reads downstream to 150 bp. The two replicates were then combined for further analysis.

COLO829 ChIP-seq data were obtained in fastq format and aligned using Burrows-Wheeler Aligner (BWA) (Li and Durbin, 2009). The resultant SAM file was converted to BAM format and sorted using SAMtools (Li et al., 2009). Peaks were called using the Homer Suite (Heinz et al., 2010) with 'factor' as the style option and IgG as the background control. Only CTCF peaks present across replicate experiments were used for analysis. To ensure standard peak sizes (332 bp by default), where peaks overlapped, peaks from the first replicate experiment were kept.

To calculate the normalised variant allele frequency (normVAF) of each COLO829 mutation from ChIP-seq data, the reads from both ChIP-seq replicate experiments were combined. Base calls at specific sites were determined using SAMtools (Li et al., 2009) mpileup on sorted BAM files of either COLO829 CTCF ChIP-seq or whole-genome sequencing (WGS) data. Reads were disregarded if they recorded a base call which was neither the wild-type nor mutant nucleotide, using base calls as previously recorded (Craig et al., 2016). Normalisation for copy number variation and mapability was performed by using COLO829 WGS read coverage to calculate the normVAF, making use of the following formula:

$$\text{normVAF} = \frac{\left(\text{Mutant}_{\text{ChIP}} \times \frac{\text{Wildtype}_{\text{WGS}}}{\text{Mutant}_{\text{WGS}}} \right)}{\left(\text{Mutant}_{\text{ChIP}} \times \frac{\text{Wildtype}_{\text{WGS}}}{\text{Mutant}_{\text{WGS}}} \right) + \text{Wildtype}_{\text{ChIP}}}$$

Repair dataset processing

Genome-wide nucleotide excision repair (NER) sequencing (excision repair sequencing; XR-seq) cyclobutane pyrimidine dimer (CPD) repair data from UV light-irradiated normal human skin fibroblast (NHF) cells (Hu et al., 2015) were obtained in Sequence Read Archive (SRA) format (GEO: GSM1659156). As previously described (Hu et al., 2015), the raw reads were trimmed using Trimmomatic (Bolger et al., 2014) and aligned using Bowtie (Langmead et al., 2009). De-duplicated aligned reads for the two replicate experiments were merged for analysis, and reads were discarded if less than 20 bp in length. Time-course CPD and (6-4)pyrimidine-pyrimidone photoproduct ((6-4)PP) data XR-seq data for NHF cells for all time points (Adar et al., 2016) were obtained in SRA format (GEO: GSE76391). Reads were processed as described above, but with reads discarded only if they were greater than 36 bp in length. This is because reads greater than this length are unlikely to be nucleotide excision fragments which are generally 22-30 nucleotides in length (Hu et al., 2015).

CTCF motif identification

The strict 13 bp CTCF motifs (n=5,470; **Table S1a**) were confined to only those motifs which adhered to the following sequence CCA(C|G)(C|T)AG(A|G)(G|T)GGC(A|G) which overlapped with any of the 150 bp merged list of peaks from the replicate CTCF ChIP-seq datasets from NHEK cells. We defined the control motif as consisting of any region overlapping a CTCF NHEK ChIP-seq peak with the same six bases at the 3' end of the motif as those in the strict 13 bp CTCF motif, but with any combination of bases at the 5' end of the motif, as follows: NNNNNNN(A|G)(G|T)GGC(A|G), where N represents any nucleotide. Control motifs were excluded if the six bases at the 3' end of the motif overlapped with any full motifs identified previously, with a final count of n=80,551 control motifs identified. A subset of n=5,470 control motifs (**Table S1b**) were randomly selected from the full list such that they were matched for replication timing with the strict CTCF motif (that is, corresponding motifs had replication timing within ± 1 of each corresponding strict CTCF motif).

To identify mutations in CTCF motifs in COLO829, motifs were identified within CTCF peaks from COLO829 CTCF ChIP-seq (only peaks common between replicate experiments were used, as described above). CTCF motifs were identified using the consensus CTCF motif from the JASPAR database (Mathelier et al., 2015), using FIMO (Grant et al., 2011) with default settings.

For the analyses of NHEK CTCF motifs, motifs were identified within CTCF peaks from NHEK CTCF ChIP-seq (the merged list of 150 bp peaks was used for this analysis, as described above). NHEK CTCF motifs (n=56,009) were identified using the consensus CTCF motif from the JASPAR database (Mathelier et al., 2015), using FIMO (Grant et al., 2011) with default settings.

- We identified CTCF motifs with RAD21 (NHEK CTCF motifs which overlapped RAD21 peaks in all eight cell-types; n=12,126) and CTCF motifs without RAD21 (NHEK CTCF motifs which did not

overlap RAD21 peaks in any cell-type; $n=6,330$). These were used for the mutation profiles at **Figure 2a** and the correlations at **Figure 2b**.

- Using the list of NHEK CTCF motifs with RAD21 (see above), we defined a subset of motifs from each group ($n=1,099$) which had matched CTCF read coverage (that is, corresponding motifs had NHEK CTCF read coverage within ± 5 reads). These motifs were used for the repair profile at **Figure 2c**.
- For the repair profile (**Figure 2d**) and correlations (**Figure 2e**), we selected from the list of NHEK CTCF motifs with RAD21, the motifs in the top (Q4) and bottom (Q1) quartiles of CTCF read coverage ($n=3,031$).

TAD anchor motifs were identified from previously published data (Rao et al., 2014) using results from Hi-C experiments in NHEK cells. For VAF calculations of mutations in anchor and non-anchor regions (**Figure S3e**), published anchor motif annotations (Rao et al., 2014) were used in addition to strict 13 bp CTCF motifs. For mutations in the strict 13 bp motif, motifs were separated into all those that fell into anchor regions ($n=1,587$; motifs within TAD anchors from NHEK cells (Rao et al., 2014)), and those that fell into non-anchor regions ($n=3,883$; all remaining motifs).

Data binning in Figure 2

Binned data is shown in **Figure 2b** and **Figure 2e**. In **Figure 2b**, for motifs with RAD21 binding, data was binned by CTCF reads across the motif, incrementally increasing by 25, with all motifs with ≥ 600 CTCF reads allocated to the final bin. For motifs without RAD21 binding, data was binned by CTCF reads across the motif, incrementally increasing by 10, with all motifs with ≥ 90 CTCF reads allocated to the final bin. In **Figure 2e**, data was allocated to seven equally-sized bins, with bins ranging from replication timing of 10 to 80 (Q1 ChIP-seq) and replication timing of 15 to 85 (Q4 ChIP-seq). Within bins, CTCF reads (**Figure 2b**) or replication timing (**Figure 2e**) were averaged to provide the y-axis value, and mutations were averaged and normalised to mutations per Mb per sample to provide the x-axis value.

Mutation simulation

Calculation of the expected distribution of mutations across CTCF binding sites utilised mutations randomly shuffled across the genome using “shuffle” from BEDtools (Quinlan and Hall, 2010), across 1,000 iterations.

Simulated mutations within the CTCF motifs were calculated by adopting trinucleotide mutation signatures as described (Alexandrov et al., 2013). Trinucleotide mutation signatures for each cancer type were determined by combining all mutations within a cancer type, and counting the number of each of the possible 96 trinucleotides. Mutations were then randomly allocated across the genome to locations with the corresponding trinucleotide, according to the number of each trinucleotide mutated within each cancer type. The mutation density was then calculated at each base of the CTCF motif using these randomly allocated mutations. This process was repeated for a total of 1,000 iterations, and the average mutation density at each base within the CTCF motif was calculated.

Bootstrapping analysis

Bootstrapping for percentage of cancer-associated genes (**Figure 4a**) was performed by selecting all TAD anchor motif mutations from melanoma samples (using TAD anchor motifs described above) and then randomly shuffling these mutations across all anchor motifs. All genes with a TSS within a gene loop with a mutated anchor motif were then identified, and the proportion of these that were cancer-associated was calculated. This process was repeated through 1,000 iterations to obtain robust confidence intervals around the mean proportion of cancer-associated genes from random iterations.

Bootstrapping for percentage of cancer-associated (**Figure S3b**) and non-cancer associated (**Figure S3c**) genes within unique anchor-gene-loop groupings was performed as follows. All TAD anchor motif mutations in melanoma samples were identified, as described above. These mutations were then randomly shuffled across all anchor motifs, and all genes with TSSs within loops with mutated anchor motifs were identified. All unique anchor-gene-loop groupings were identified, and expression differences from melanoma RNA-sequencing data were calculated as described above, using wild-type and mutant melanoma sample designations from the randomly shuffled mutations. Gene expression differences were identified either by unpaired *t*-test or unpaired one-sample *t*-test, whichever was appropriate. Multiple testing correction was then performed using the Bonferroni correction method on all *P*-values obtained, and the proportion of significantly differentially-expressed genes within each iteration after multiple testing correction was determined, using a threshold of $P <$

0.05 to determine significance. A distribution of results was calculated by repeating this process through 1,000 iterations.

Supplemental References

Only additional references not cited in the main manuscript are listed here.

Bolger, A.M., Lohse, M., and Usadel, B. (2014). Trimmomatic: a flexible trimmer for Illumina sequence data. *Bioinformatics* 30, 2114-2120.

Grant, C.E., Bailey, T.L., and Noble, W.S. (2011). FIMO: scanning for occurrences of a given motif. *Bioinformatics* 27, 1017-1018.

Langmead, B., Trapnell, C., Pop, M., and Salzberg, S.L. (2009). Ultrafast and memory-efficient alignment of short DNA sequences to the human genome. *Genome Biol* 10, R25.

Li, H., Handsaker, B., Wysoker, A., Fennell, T., Ruan, J., Homer, N., Marth, G., Abecasis, G., and Durbin, R. (2009). The Sequence Alignment/Map format and SAMtools. *Bioinformatics* 25, 2078-2079.

Quinlan, A.R., and Hall, I.M. (2010). BEDTools: a flexible suite of utilities for comparing genomic features. *Bioinformatics* 26, 841-842.